🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 2 · STANDARD 6.DS.3

Describe Data by Range and Interquartile Range

Lesson 2-5 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes
Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the range and the interquartile range of a data set and use them to describe how spread out the data is.

Language Objective: I can explain what the spread of a data set is using the words range, quartile, interquartile range, and vary.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Range and Interquartile Range | Range | Quartile | Interquartile range | Variability | Measure of variation

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The range uses the ______ values; the interquartile range uses the ______.
2. Subtract the least value from the ______ value to find the range.
3. Numbers that split ordered data into four equal parts are ______.
4. The spread of the middle half of the data, Q3 – Q1, is the ______ ______.

5 How spread out the numbers in a data set are is its ______.

6 A single number describing how spread out a data set is is a ______ ______ ______.

Write About the Math

Which class scored better overall, and how do the range and the IQR show it?
¿A qué clase le fue mejor en general, y cómo lo muestran el rango y el rango intercuartílico?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Range and Interquartile Range** *Rango y rango intercuartílico*
- ☐ **Range** *Rango*
- ☐ **Quartile** *Cuartil*
- ☐ **Interquartile range** *Rango intercuartílico*
- ☐ **Variability** *Variabilidad*

Model: *Equal medians do not make two data sets the same — you still need to know how spread out the scores are.* — Students name a measure of spread rather than another measure of center.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Class A's range is ____ and Class B's range is ____, so Class A's scores are more ____. Class A's IQR is ____ and Class B's IQR is ____, so the middle half of Class B is packed into a ____ stretch of points. Because 75% of Class B scored ____ or more, Class B did better ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**

Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Range and Interquartile Range
2. **Notes 2:** Range
3. **Notes 3:** Quartile
4. **Notes 4:** Interquartile range
5. **Notes 5:** Variability
6. **Notes 6:** Measure of variation
7. **Watch for this mistake:** *A common mistake with Range and Interquartile Range is adding the two quartiles instead of subtracting them — writing IQR = Q3 + Q1. With Q1 = 42 and Q3 = 58 that gives 100, which is larger than the data set's entire range of 20. Every measure of variation is a DIFFERENCE, and the IQR can never exceed the range, because the middle half is part of the whole span. If your IQR comes out bigger than your range, you added when you should have subtracted.*
8. **Try It 1:** *— Range and IQR are both measures of variation; a quartile is a marker, and variability is the idea both measures describe.*
9. **Try It 2:** A. B, because 25 − 5 = 20 *— A's range is 18 − 12 = 6. B's range is 25 − 5 = 20, so B is more spread out.*